DeadlineDinosaur: Fast Gaussian Splatting for SIGGRAPH Asia's 3D Gaussian Splatting Challenge

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1 Introduction

SIGGRAPH Asia 2025 presents a new challenge: the 3D Gaussian Splatting Challenge focused on High-Quality and Fast 3DGS Reconstruction. This challenge seeks to advance efficient 3D Gaussian Splatting (3DGS) reconstruction methodologies capable of producing high-quality results with minimal latency. The competition mandates that participating systems complete the entire 3DGS reconstruction pipeline within 60 seconds on a single NVIDIA GeForce RTX 4090 Ti GPU running Ubuntu 22.04, thereby emphasizing the critical balance between reconstruction fidelity and computational efficiency.

The challenge dataset encompasses diverse and complex scenes, including full 360-degree object captures and partial coverage scene captures. These varied scenarios test the robustness and adaptability of reconstruction algorithms across different imaging conditions and scene complexities.

1.1 3D Gaussian Splatting

3D Gaussian Splatting has emerged as a highly efficient approach for novel view synthesis and 3D scene reconstruction. In contrast to traditional neural rendering pipelines that rely on implicit scene representations, 3DGS models scenes explicitly using anisotropic 3D Gaussian primitives and renders them through efficient rasterization. This explicit structure enables rapid optimization and real-time rendering, which makes 3DGS well suited for time-constrained and latency-sensitive applications.

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The development of fast 3DGS reconstruction methods is particularly relevant for edge computing and mixed reality platforms. With the increasing availability of powerful consumer headsets such as the Apple Vision Pro, the Meta Quest series, and the Samsung Galaxy XR, there is a growing demand for on-device 3D reconstruction. Achieving high-quality 3DGS reconstruction within strict time and resource budgets would support real-time spatial mapping, immersive content creation, and dynamic scene understanding directly on these devices, thereby reducing dependence on cloud infrastructure and lowering end-to-end latency for interactive mixed reality experiences.

1.2 Recent Works in Efficient 3DGS

Several recent innovations address the challenge of enabling high-fidelity 3DGS reconstruction under limited computational budgets. A first line of work focuses on reducing model size by controlling the growth of Gaussian primitives. Taming 3DGS [4] addresses the problem of excessive memory consumption through a guided constructive densification process that constrains the final model size to a predefined budget. By employing score-based densification to ensure that each added Gaussian meaningfully contributes to reconstruction quality, this approach achieves reductions of four to five times in both model size and training time. Complementing model-size control, Speedy-Splat [2] accelerates both rendering and optimization by improving the rendering pipeline. It introduces the SnugBox and AccuTile algorithms for precise Gaussian localization and incorporates a compression strategy that combines Soft Pruning with Hard Pruning to remove redundant primitives. These improvements lead to an average rendering speedup of 6.71 times.

A second line of work focuses on accelerating optimization. DashGaussian [1] improves training efficiency by managing computational complexity through a dynamic rendering resolution scheme and a primitive-growth scheduler. This formulation treats optimization as a process of progressively fitting higher-frequency components of the scene, thereby reducing unnecessary computation in early iterations. The method achieves an average training-speed improvement of 45.7 percent across several 3DGS backbones and produces high-quality reconstructions in under 200 seconds. Finally, LiteGS [3] proposes a high-performance framework that reduces training cost through coordinated system-level and algorithmic design. It integrates a hardware-aware warp-based raster at the low level, a Cluster–Cull–Compact pipeline to improve data locality at the mid level, and a densification criterion based on the variance of the opacity gradient at the algorithmic level. Together, these components accelerate standard 3DGS training by as much as 13.4 times and establish a new efficiency benchmark for high-quality scene reconstruction.

2 Method

Our method builds upon **LiteGS** [3] as the primary training backbone and incorporates a resolution-scaling strategy inspired by the frequency-aware scheduling introduced in **DashGaussian** [1]. This combination maintains LiteGS's system-level efficiency while enabling a more structured and computationally adaptive optimization process.

LiteGS provides a strong foundation for accelerated 3D Gaussian Splatting through a coordinated design across system, memory, and algorithmic layers. At the system level, its hardware-aware warp-based rasterizer exploits warp-level parallelism to substantially reduce the cost of Gaussian projection. Complementing this, the Cluster–Cull–Compact pipeline improves memory locality and bandwidth efficiency by spatially clustering Gaussians, culling those irrelevant to a given view, and compacting the active set during each iteration. At the algorithmic level, LiteGS adopts the score-based densification criterion from Taming3DGS, which ensures that new Gaussians are added only when they provide meaningful improvements to reconstruction quality. This controlled approach to densification eliminates redundant primitive growth while maintaining high fidelity, making LiteGS particularly suitable for time-constrained reconstruction.

While LiteGS achieves substantial acceleration through these system and algorithmic innovations, it operates at a fixed full rendering resolution across the entire optimization process. To further improve efficiency, we integrate the resolution-scaling principle introduced by DashGaussian. DashGaussian formulates 3DGS optimization as a progression from fitting low-frequency content to progressively higher-frequency details. This insight motivates a dynamic rendering-resolution schedule in which training begins at a low spatial resolution, allowing the model to capture coarse global structure at minimal computational cost. As optimization stabilizes, the rendering resolution is gradually increased to enable the reconstruction of mid- and high-frequency scene details.

We adopt this scheduling logic within LiteGS to modulate rendering cost in accordance with the evolving frequency complexity of the scene. By avoiding unnecessary high-resolution rasterization in the early iterations, the combined framework reduces overall computation while preserving the ability to recover fine details in later stages. This synergy between LiteGS's system-aware acceleration and DashGaussian's frequency-guided resolution scaling yields a highly efficient optimization pipeline capable of producing high-quality reconstructions within strict time constraints. accelerate training.

3 Results

Our method successfully reconstructs all challenge scenes within the 60-second time constraint while maintaining competitive visual quality. Figure 1 presents the scene-wise PSNR values, demonstrating consistent performance across diverse scene types. Figure 2 shows qualitative comparisons between ground truth and our rendered results, illustrating the visual fidelity achieved by our approach. Table 1 summarizes the quantitative performance metrics.

Table 1: Final reconstruction metrics across all scenes.

Metric	Value
Average PSNR	25.05
Average Time (s)	56.92

4 Conclusion

Our final framework combines LiteGS's system-aware acceleration with DashGaussian's frequency-guided resolution scaling. Training begins at a low resolution and progressively increases to the full target resolution, while LiteGS's score-based densification and optimized rasterizer efficiently manage Gaussian growth and projection throughout the process. This hybrid strategy maintains reconstruction fidelity while providing substantial speedups, enabling high-quality 3D reconstruction within the strict 60-second constraint.

We have presented *DeadlineDinosaur*, our solution to the SIGGRAPH Asia 2025 3D Gaussian Splatting Challenge. By augmenting LiteGS with DashGaussian's frequency-aware optimization schedule, our method achieves high-quality reconstruction within the 60-second deadline on a single RTX 4090 Ti GPU. The results demonstrate that carefully designed scheduling of computational complexity during optimization yields an effective balance between reconstruction quality and execution time. Our approach contributes toward making 3D Gaussian Splatting practical for real-time and edge-computing scenarios, particularly in the context of emerging mixed reality platforms.

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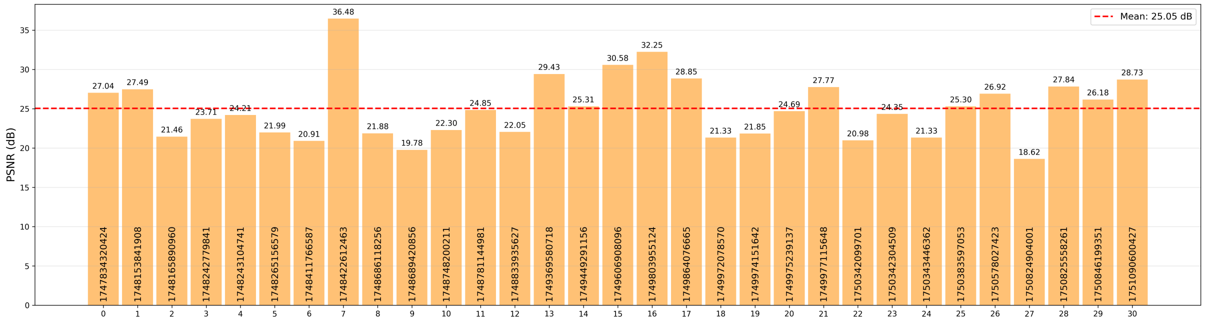
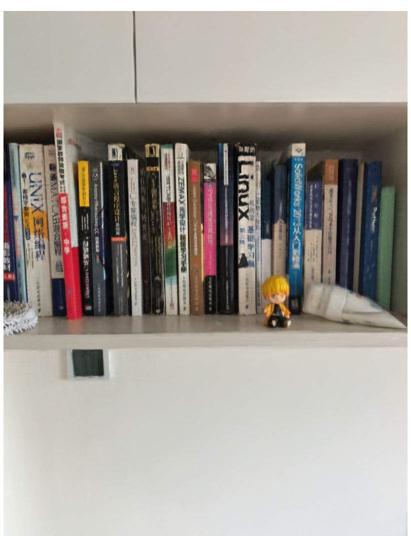


Figure 1: Scene-wise PSNR comparison across the challenge dataset. The red dotted-line indicates the average PSNR.

(a) Ground Truth - Scene 1

(b) Ground Truth - Scene 2

(c) Ground Truth - Scene 3



(d) Our Render - Scene 1

(e) Our Render - Scene 2

(f) Our Render - Scene 3

Figure 2: Qualitative comparison of ground truth and our rendered results for selected scenes from the challenge dataset.